# Lecture 22: Section 5.9

## u-Substitution

Md Ahsanul Haque, Ph.D.

Assistant Professor
Department of Mathematics and Physics
North South University, Dhaka

# The Substitution Theorem

## Theorem (Substitution Rule for Definite Integrals)

*If $g'(x)$ is continuous on $[a, b]$ and $f$ is continuous on an interval containing $g([a, b])$, then*

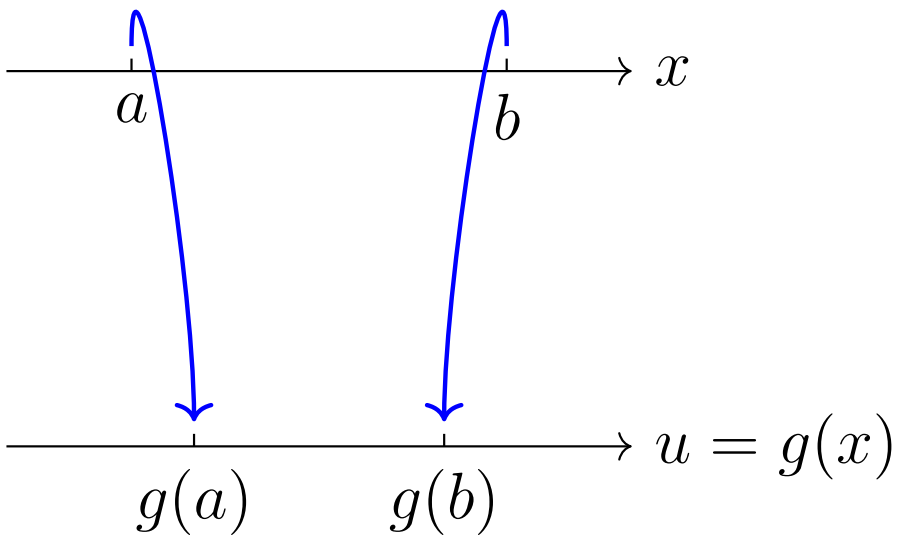
$$\int_a^b f(g(x)) \cdot g'(x)\, dx = \int_{g(a)}^{g(b)} f(u)\, du$$

*where $u = g(x)$.*

## What This Means

1. Change the limits of integration to $u$-limits and never go back to $x$.
2. Keep the $x$-limits, integrate in $u$, then convert back to $x$ before evaluating.

# Why This Works: Visual Intuition




**Key Idea:** The substitution $u = g(x)$ transforms the interval $[a, b]$ in $x$ to the interval $[g(a), g(b)]$ in $u$.

# Method 1: Keep Original Limits

## Step-by-Step Procedure

1. Choose $u = g(x)$ and compute $du = g'(x)dx$.
2. **Keep original $x$-limits** $a$ and $b$.
3. Rewrite integrand in terms of $u$ (treat as indefinite integral).
4. Find antiderivative $F(u)$.
5. Convert back to $x$: $F(g(x))$.
6. Evaluate $F(g(x))$ from $x = a$ to $x = b$.

## When to Use This Method

- When $g(x)$ is messy or hard to invert.
- When you're more comfortable with indefinite integrals.
- When the substitution doesn't simplify the limits much.

# Method 2: Change Limits (Recommended)

## Step-by-Step Procedure

1. Choose $u = g(x)$ and compute $du = g'(x)dx$.
2. **Change the limits:**
   - Lower limit: When $x = a$, $u = g(a)$
   - Upper limit: When $x = b$, $u = g(b)$
3. Rewrite the entire integral in terms of $u$ with the new limits.
4. Evaluate the $u$-integral.
5. Answer is in terms of $u$; no need to return to $x$.

## Advantage

- Cleaner: Answer is a number directly.
- No chance of forgetting to convert back to $x$.
- More efficient for complex substitutions.

# Comparison of the Two Methods

**Method 1: Keep Limits**

$$\int_a^b f(g(x))g'(x)dx$$

$\Downarrow$ Find $F(u) = \int f(u)du$

$\Downarrow$ $F(g(x))\Big|_{x=a}^{x=b}$

$\Downarrow$ Evaluate

**Number**

**Method 2: Change Limits**

$$\int_{x=a}^{x=b} f(g(x))g'(x)dx$$

$\Downarrow$ Substitute $u = g(x)$

$$\int_{u=g(a)}^{u=g(b)} f(u)du$$

$\Downarrow$ Evaluate

**Number**

# Example 2: Polynomial Substitution

## Example

Evaluate $\int_0^2 3x^2\sqrt{x^3+1}\,dx$.

**Solution using Method 2:**

1. Let $u = x^3 + 1$, so $du = 3x^2dx$.
2. **Change limits:**
   - When $x = 0$: $u = 0^3 + 1 = 1$
   - When $x = 2$: $u = 2^3 + 1 = 9$
3. Rewrite: $\int_0^2 3x^2\sqrt{x^3+1}\,dx = \int_{u=1}^{u=9}\sqrt{u}\,du$
4. Evaluate: $\int_1^9 u^{1/2}\,du = \left[\frac{2}{3}u^{3/2}\right]_1^9$
5. Compute: $\frac{2}{3}(9^{3/2} - 1^{3/2}) = \frac{2}{3}((2^3)^{3/2} - 1) = \frac{2}{3}(2^{9/2} - 1)$
6. Simplify: $9^{3/2} = (\sqrt{9})^3 = 27$
   so answer $= \frac{52}{3}$

# Example 2: Trigonometric Substitution

## Example

Evaluate $\displaystyle\int_0^{\pi/4} \sec^2 x \tan x \, dx.$

**Solution using Method 2:**

1. Let $u = \tan x$, so $du = \sec^2 x \, dx$.
2. **Change limits:**
   - When $x = 0$: $u = \tan 0 = 0$
   - When $x = \pi/4$: $u = \tan(\pi/4) = 1$
3. Rewrite: $\displaystyle\int_0^{\pi/4} \sec^2 x \tan x \, dx = \int_{u=0}^{u=1} u \, du$
4. Evaluate: $\displaystyle\int_0^1 u \, du = \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2} - 0 = \frac{1}{2}$

# Example 3: Exponential Substitution

## Example

Evaluate $\displaystyle\int_0^1 \frac{e^x}{1+e^x}\,dx.$

**Solution using Method 2:**

1. Let $u = 1 + e^x$, so $du = e^x dx$.
2. **Change limits:**
   - When $x = 0$: $u = 1 + e^0 = 1 + 1 = 2$
   - When $x = 1$: $u = 1 + e^1 = 1 + e$
3. Rewrite: $\int_0^1 \frac{e^x}{1+e^x}\,dx = \int_{u=2}^{u=1+e} \frac{1}{u}\,du$
4. Evaluate: $\int_2^{1+e} \frac{1}{u}\,du = [\ln|u|]_2^{1+e}$
5. Compute: $\ln(1+e) - \ln 2 = \ln\left(\frac{1+e}{2}\right)$

# Example 4: Using Method 1

## Example

Evaluate $\int_0^1 2x(x^2+3)^4\,dx$ using Method 1.

**Solution using Method 2:**

1. Let $u = x^2 + 3$, so $du = 2xdx$.
2. **Keep original limits** (0 and 1 for now).
3. Find antiderivative: $\int 2x(x^2+3)^4 dx = \int u^4 du = \frac{u^5}{5} + C$
4. Convert back to $x$: $\frac{(x^2+3)^5}{5}$
5. Evaluate with original limits:

$$\left[\frac{(x^2+3)^5}{5}\right]_0^1 = \frac{(1^2+3)^5}{5} - \frac{(0^2+3)^5}{5} = \frac{4^5}{5} - \frac{3^5}{5} = \frac{781}{5}$$

# Example 5: Comparing Both Methods

## Example

Evaluate $\int_1^2 \frac{(\ln x)^2}{x}\,dx$ using both methods.

**Method 1: Keep limits**

1. $u = \ln x,\ du = \frac{1}{x}dx$
2. Antiderivative: $\int u^2 du = \frac{u^3}{3} + C = \frac{(\ln x)^3}{3} + C$
3. Evaluate: $\left[\frac{(\ln x)^3}{3}\right]_1^2 = \frac{(\ln 2)^3}{3} - \frac{(\ln 1)^3}{3} = \frac{(\ln 2)^3}{3} - 0 = \frac{(\ln 2)^3}{3}$

**Method 2: Change limits**

1. $u = \ln x,\ du = \frac{1}{x}dx$
2. Limits: $x = 1 \Rightarrow u = \ln 1 = 0,\ x = 2 \Rightarrow u = \ln 2$
3. $\int_1^2 \frac{(\ln x)^2}{x}dx = \int_0^{\ln 2} u^2 du = \left[\frac{u^3}{3}\right]_0^{\ln 2} = \frac{(\ln 2)^3}{3}$

# Example 6: When Substitution Isn't Obvious

## Example

Evaluate $\int_0^{\pi/2} \sin^3 x \cos x \, dx$.

**Solution:**

1. Let $u = \sin x$, so $du = \cos x dx$
2. Limits: $x = 0 \Rightarrow u = \sin 0 = 0$, $x = \pi/2 \Rightarrow u = \sin(\pi/2) = 1$
3. Rewrite: $\int_0^{\pi/2} \sin^3 x \cos x dx = \int_0^1 u^3 du$
4. Evaluate: $\left[\frac{u^4}{4}\right]_0^1 = \frac{1}{4} - 0 = \frac{1}{4}$

**Alternative without substitution:** We could recognize that $\frac{d}{dx}(\sin^4 x) = 4 \sin^3 x \cos x$, so

$$\int_0^{\pi/2} \sin^3 x \cos x dx = \frac{1}{4} \left[\sin^4 x\right]_0^{\pi/2} = \frac{1}{4}(1 - 0) = \frac{1}{4}$$

# Warning: The Substitution Must Be One-to-One

## Important Condition

The substitution theorem requires $g'(x)$ to be continuous. More importantly, if $g$ is not one-to-one on $[a, b]$, Method 2 can fail!

## Example (Problematic Case)

Consider $\int_{-1}^{1} x^2 dx$ with substitution $u = x^2$.

- When $x = -1$, $u = 1$; when $x = 1$, $u = 1 \Rightarrow$ limits are the same!
- But $\int_{-1}^{1} x^2 dx = \frac{2}{3} \neq 0$.
- Problem: $u = x^2$ is not one-to-one on $[-1, 1]$.

For Method 2, ensure $g$ is one-to-one on $[a, b]$, or split the integral where $g$ is one-to-one.

# Dealing with Non-One-to-One Substitutions

## Example

Evaluate $\int_{-1}^{1} 2x\sqrt{x^2+1}dx$.

**Solution:** $u = x^2 + 1$, $du = 2xdx$

- The substitution $u = x^2 + 1$ is not one-to-one on $[-1, 1]$.
- But note: $f(x) = 2x\sqrt{x^2+1}$ is **odd** $(f(-x) = -f(x))$.
- Symmetry: $\int_{-1}^{1} 2x\sqrt{x^2+1}dx = 0$.

**Alternative:** Use Method 1:

1. Antiderivative: $\int 2x\sqrt{x^2+1}dx = \frac{2}{3}(x^2+1)^{3/2} + C$
2. Evaluate: $\left[\frac{2}{3}(x^2+1)^{3/2}\right]_{-1}^{1} = \frac{2}{3}(2^{3/2}) - \frac{2}{3}(2^{3/2}) = 0$

**Lesson:** When $g$ isn't one-to-one, consider symmetry or use Method 1.

# Practice Problems: Basic Substitutions

Evaluate using substitution (choose either method):

$$\int_0^1 xe^{x^2}\,dx$$

$$\int_1^2 \frac{dx}{x\ln x}$$

$$\int_0^\pi \sin x\cos^2 x dx$$

$$\int_0^1 \frac{x}{\sqrt{x^2+9}}dx$$

$$\int_0^{\pi/4} \tan x\sec^2 x dx$$

# Practice Problems: More Challenging

Evaluate $\int_0^1 x(1-x)^4 dx$ using $u = 1 - x$.

Evaluate $\int_0^{\ln 3} \frac{e^x}{e^x + 2} dx$.

Evaluate $\int_1^e \frac{\ln x}{x} dx$ using both methods.

Find $\int_0^2 x\sqrt{x+2} dx$ using $u = x + 2$.

Evaluate $\int_{-1}^1 \frac{x}{x^2 + 1} dx$ without substitution.

# Practice Problems: Theory and Concepts

True or False:

- When using $u$-substitution in a definite integral, you must always change the limits of integration.
- If $\int_1^4 f(x)dx = 7$ and we substitute $u = 2x$, what are the new limits for $\int f(2x)dx$?
- Explain in your own words why Method 2 (changing limits) is generally preferred over Method 1.
- Give an example of a substitution where Method 2 would fail because the substitution function is not one-to-one.
- If $\int_0^3 g(x)dx = 5$ and we let $u = g(x)$, what is $\int_{g(0)}^{g(3)} udu$? (Careful!)

# Method 1: Keep Original Limits

1. **Substitute:** $u = g(x)$, $du = g'(x)dx$
2. **Keep $x$-limits** $a$ and $b$
3. **Find antiderivative** as an indefinite integral in $u$
4. **Convert back** to $x$: $F(g(x))$
5. **Evaluate** $F(g(x))$ from $x = a$ to $x = b$

# Method 2: Change Limits (Usually Better)

1. **Substitute:** $u = g(x)$, $du = g'(x)dx$
2. **Change limits:** $x = a \to u = g(a)$, $x = b \to u = g(b)$
3. **Rewrite:** $\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$
4. **Evaluate** the $u$-integral
5. **Answer** is a number directly

# Important Reminders

## When to Use Which Method

- **Method 1:** When $g$ is not one-to-one, or when you're more comfortable with indefinite integrals.
- **Method 2:** When $g$ is one-to-one on $[a, b]$ and $g(a), g(b)$ are easy to compute.

## Pitfalls to Avoid

- Forgetting to change limits in Method 2 (or changing them incorrectly).
- Forgetting to convert back to $x$ in Method 1.
- Using a substitution that's not one-to-one on the entire interval.
- Not checking that $g'(x)$ is continuous on $[a, b]$.

# Visual Summary

